

Exploring the Music Genre Space Using Voronoi and Delaunay Diagrams

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Abstract

In this study, I explore the relationships between musical artists based on their audio features by using Voronoi diagrams and Delaunay triangulation. I analyze tracks from a dataset containing multiple artists and map them in a 2D space defined by *energy* and *acousticness*. This approach allows me to visualize both the influence regions of each artist and the proximity between artists with similar musical characteristics. My findings demonstrate that geometric visualization techniques provide intuitive insights into music analysis and artist similarity.

1 Introduction

Understanding musical similarity is a fundamental challenge in music analysis, recommendation systems, and playlist curation. While features such as genre, tempo, or popularity can provide some insight, they often fail to capture the nuanced differences between artists. I wanted to investigate whether geometric visualization techniques like Voronoi diagrams and Delaunay triangulation could offer a clearer understanding of how artists relate to one another in feature space.

The goal of my study was to create a visual representation that shows not only the position of each artist in a feature space but also their “area of influence” and their proximity to similar artists. By using this approach, I aim to identify clusters of similar artists, outliers with unique characteristics, and potential patterns in the music space.

2 Model and Approach

I focused on two key audio features from the dataset:

- **Energy:** This feature measures the intensity and activity level of a track, from 0 (very calm) to 1 (highly energetic). It captures the overall dynamism of a song.
- **Acousticness:** This represents the likelihood that a track is acoustic, from 0 (not acoustic) to 1 (fully acoustic). It helps distinguish between electronic and acoustic music styles.

The dataset I used contains multiple tracks from a large number of artists. To ensure the visualization would be clear and computationally feasible, I decided to:

1. Select the top 10 artists by number of tracks available in the dataset.

2. Compute the mean *energy* and *acousticness* for each artist.
3. Convert these averaged values into 2D points for visualization in the energy-acousticness space.

2.1 Voronoi Diagram

I employed Voronoi diagrams to partition the feature space. Each region corresponds to an artist, containing all points that are closer to that artist than to any other. This allows me to visualize the “influence” of each artist and understand which artists are similar or dominant in specific areas of the feature space.

2.2 Delaunay Triangulation

The Delaunay triangulation complements the Voronoi diagram by connecting points into triangles such that no other points lie inside the circumcircle of any triangle. This triangulation helps highlight relationships and clusters between artists, showing which artists are direct neighbors in the feature space.

3 Experiment and Visualization

I implemented the analysis in Python, using `pandas` for data handling, `matplotlib` for plotting, and `scipy.spatial` for computing Voronoi and Delaunay structures. After selecting the top 10 artists, I calculated the average energy and acousticness per artist and generated a combined plot showing both the Voronoi diagram and the Delaunay triangulation.

In the visualization:

- Red points represent the artists.
- Voronoi regions outline the area of influence for each artist.
- Gray dashed lines show Delaunay triangles.
- Labels next to each point indicate the artist name.

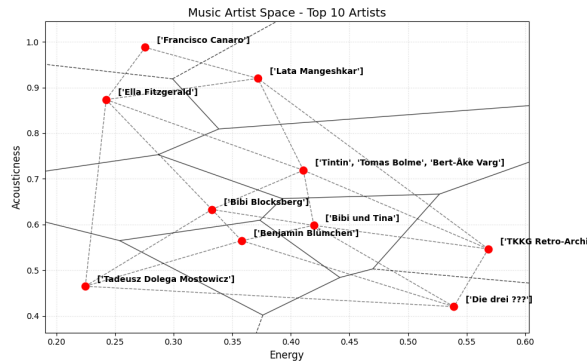


Figure 1: Voronoi and Delaunay visualization of top 10 artists in energy-acousticness space.

4 Results and Discussion

From my visualization, I observed that artists with similar audio characteristics are located near each other, with adjacent Voronoi regions. Delaunay triangles further emphasize these relationships, showing which artists share direct connections in the feature space. Some artists occupy distinct regions, indicating that they have unique musical characteristics compared to others in the dataset.

This method provides several insights:

- **Cluster Identification:** Groups of similar artists are easily identifiable.
- **Outlier Detection:** Artists that do not share characteristics with the main clusters stand out clearly.
- **Feature Relationships:** The combined analysis of energy and acousticness gives a simple but effective way to understand how features interact.

I also realized that expanding this approach to include additional audio features, such as *danceability*, *valence*, or *tempo*, could provide a higher-dimensional understanding of artist relationships. In such cases, dimensionality reduction techniques like PCA could help visualize the data in 2D or 3D.

5 Conclusion

Through this project, I learned that Voronoi diagrams and Delaunay triangulation can be powerful tools for visualizing relationships between musical artists based on audio features. By focusing on top artists and averaging their feature values, I was able to generate clear and interpretable visualizations. This approach could be used in music recommendation systems, playlist generation, or broader musicological studies. Overall, geometric visualization techniques provide an intuitive and visually appealing method to explore musical similarities and differences.